

SUMANTH REDDY KASIREDDY

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[Portfolio](#) | [LinkedIn](#) | [GitHub](#)

PROFESSIONAL SUMMARY

Full-stack developer specializing in AI-integrated, animation-rich web applications, with hands-on expertise using React.js, Node.js, REST APIs, Supabase (PostgreSQL), Three.js, and GSAP. Strong foundation in Data Structures, Algorithms, and System Design principles, backed by consistent competitive programming in C++. Passionate about building highly scalable, secure, and visually captivating digital products that solve real-world problems.

TECHNICAL SKILLS

Languages: C++, C, JavaScript (ES6+), SQL

Frameworks & Libraries: React.js, Node.js, Express.js, REST APIs, Three.js (React Three Fiber), GSAP, Tailwind CSS, Canvas API

AI & Cloud Services: Gemini Pro / Vision API, Supabase (PostgreSQL), Vercel, Render

Tools & Practices: Git, GitHub, JWT / Token Authentication, Data Structures & Algorithms

FEATURED PROJECTS

Interactive 3D Personal Portfolio

2026

React.js | Three.js (React Three Fiber) | GSAP | Tailwind CSS | EmailJS | Vercel

- **Interactive 3D Graphics:** Built immersive 3D visuals utilizing React Three Fiber, including a mechanical keyboard hero model and an interactive rotating globe model, with dynamic ambient lighting.
- **Scroll-Driven Animations:** Implemented fluid micro-interactions and scroll-triggered motion effects (GSAP SplitText, BorderGlow carousel) across Hero, About, Work, and Contact sections.
- **RAG-Based AI Assistant:** Engineered a RAG-based AI assistant using Jina AI embeddings and Supabase pgvector for semantic search, with Groq for real-time LLM inference.

Plantio — AI-Powered Botanical & Plant Care Platform

2026

React.js | Node.js | Gemini AI Vision API | Supabase (PostgreSQL) | REST APIs | HTML5 Canvas API | Vercel | Render

- **AI Disease Diagnosis & Botanical Scanner:** Engineered an AI Doctor gateway utilizing Gemini Vision API to analyze plant leaf images and text queries in real time, detecting 740+ plant diseases, pests, and nutrient deficiencies with organic remedy suggestions.
- **Interactive Site-Wide AI Chatbot:** Built a floating, responsive AI assistant modal (AIChatModal) available across all pages for instant interactive plant troubleshooting, context-aware care advice, and botanical Q&A.
- **Photo Growth Journal & Canvas Compression:** Developed a multi-stage visual growth timeline (Day 1 to Year 1) with client-side HTML5 Canvas image compression (JPEG 0.7, max 450px), optimizing photo uploads to under 30KB for instant cloud sync to Supabase PostgreSQL.
- **Smart Watering Tracker & Batch Operations:** Implemented dynamic watering status badges (Watered, Due Today, Overdue) with 1-click single and batch "Water All" actions, syncing custom care frequencies across user devices.
- **Full-Stack REST Architecture & 300K+ Species DB:** Built a standalone Node.js REST backend with HMAC token auth and integrated Wikipedia & botanical APIs to serve real-time search, category filters, and detailed care guides across 300,000+ plant species.

Competitive Programming & Algorithmic Problem Solving

2025 – Present

C++ | Data Structures & Algorithms | LeetCode | CodeChef

- **Algorithmic Breadth:** Solved 100+ algorithmic problems spanning Arrays, Strings, Dynamic Programming, Binary Search, Graph Traversals, and String Matching (e.g., Rotated Sorted Array Search, House Robber, Regular Expression Matching).

EDUCATION

Bachelor of Science in Computer Science

Expected 2027

University of Massachusetts Lowell, Lowell, MA

Bachelor of Technology in Computer Science & Engineering

2023 – 2026

SR International Institute of Technology, Hyderabad

Intermediate Certificate (MPC)

April 2023

Narayana Junior College, Madhapur, Hyderabad